CLIMATE-INDUCED SALINITY AND COASTAL VULNERABILITY IN KHULNA DIVISION: POLICY OPTIONS FOR SUSTAINABLE DEVELOPMENT

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Key Points :

- Khulna Division is located on Bangladesh's climate-exposed southwest border, where severe salinity intrusion into rivers, soils, and groundwater is being driven by sea level rise, frequent cyclones, and changes in upstream flow. This has created a complicated environmental and development crisis by turning an agricultural landscape that was primarily based on rice into one that is dominated by saline soils, shrimp ponds, and soggy polders. Coastal districts like Khulna, Satkhira, and Bagerhat are where the ensuing pressures are concentrated.

- Salinity and cyclones together undermine food security, health, water supply and livelihoods. Farmers face declining yields. They are pushed toward shrimp monoculture, which can worsen soil and canal salinity and reduce rural employment. At the same time, saline contamination of ponds and tube wells has produced a chronic drinking-water crisis, with women travelling long distances for safe water and communities, experiencing rising hypertension, pregnancy complications and waterborne diseases.

- The brief argues that no single intervention is enough. Khulna requires an integrated package of responses. It can include scaling up salt-tolerant and diversified agriculture, shifting from monoculture shrimp to mixed agro-aquatic systems, and investing in community-managed safe water and climate-resilient embankments. Alongside this, targeted social protection, gender-responsive planning and stronger government actions are essential to move the division from repeated crisis management toward long-term climate resilience and inclusive development.

Introduction

The Khulna Division includes coastal districts like Khulna, Satkhira, and Bagerhat. It is situated at the most climate-exposed edge of Bangladesh's southwest delta and it is home to over 17 million people. Because of its location along the Bay of Bengal and within the lower Ganges–Brahmaputra–Meghna system, it is particularly vulnerable to storm surges, cyclones, sea level rise, and changes in upstream river flows. Over the past few decades, these processes have increased saltwater intrusion into rivers, canals, groundwater, and agricultural land, transforming the area from a predominantly rice-based agricultural area to a

patchwork of saline soils, shrimp ponds, and soggy polders.

This policy brief identifies climate-induced salinity intrusion and associated coastal vulnerability as a major structural challenge in Khulna Division. It explains how salinity, cyclones and waterlogging affect agriculture, drinking water, health and livelihoods, and how these interlinked problems hinder development in both the division and Bangladesh as a whole. The brief argues that addressing salinity with urgency, while ensuring long-term sustainability. It is essential for meeting national food security, poverty reduction and climate-resilience goals, as laid out in the Bangladesh Delta Plan 2100. It then proposes evidence-based policy options that build on successful adaptation practices in coastal Bangladesh and other deltas and adapts them to Khulna's specific ecological and socioeconomic growth.

State of the Problem

I. Salinity intrusion and changing landscapes

Bangladesh-one of the most climate-vulnerable countries in the world as it sits in a lowlying landmass that is a delta of three main rivers: the Ganges, Brahmaputra, and Meghna; is densely populated, and with natural resources as the heart of the country (Halimuzzaman et al., 2024). Salt water intrusion is one of several environmental challenges. Salinity rise is a serious risk, particularly in the South West coastal region where millions of people depend on for their lives and livelihoods (Mokbul Morshed Ahmad, 2021). Salinity intrusion relates to the movement of saline water into freshwaters such as rivers, soils and aquifers, causing physical and chemical changes (Robert Maliva, 2020). This has been speeding up in Bangladesh due to a range of natural processes such as sea level rise

and tidal surges and human induced pressures like diversion of freshwater at the headwaters, unsustainable land use practices, and unregulated shrimp farming. The coastal region of Bangladesh consists of 19 districts where around 35 million people live and their livelihood predominantly relies on agriculture, aquaculture, and small-scale fishing (Atiqur Rahman Sunny, 2020) . In recent decades, the saline front has advanced toward the head of the estuary, particularly during dry periods when low river runoff fails to drive seawater back (Hoi Soo Jung, 2023). In districts such as Khulna, Satkhira, and Bagerhat, for example, surface water and groundwater have become significantly more saline, such that there are many sources of drinking water and irrigation that can no longer be tapped.

Source : https://www.geojournal.net/uploads/archives/7-6-8-277.pdf

II. Cyclones, waterlogging and displacement

Dacope Upazila of Khulna district was selected as the study area to map out the assessment on climate change adaptation practices in order to mitigating climatic risks. The study was performed in Chalna and Kamarkhola Union. Dacope upazila is adjacent with the Sundarban mangrove forest of Khulna district (Bagchi et al., 2019). It is bounded by Pasur River on the south, Batiaghata upazila on the north, Mongla and Rampal Upazila of Bagerhat district on the east, Koyra and Paikgachha Upazila of Satkhira district on the west.

Factors considered for study area selection - Following factors were considered for the selection of the study area:

- Higher degree of risk due to climate change related hazards like cyclone, salinity intrusion, river erosion, tidal surge and coastal inundation, etc.
- Availability of adaptation program or project to combat against climate change both by GOs and NGOs.

Source : https://www.universepg.com/public/img/storage/journal-pdf/Assessment%20on%20climate%20change%20adaptation_a%20study%20on%20coastal%20area.pdf

III. Drinking water crisis and health impacts

Natural drinking water sources, such as rivers and groundwater have been threatened by saltwater intrusion from the Bay of Bengal (Talukder, 2016). Furthermore, groundwater, cultivable lands, and many rivers are being increasingly affected by saline water from higher tidal waves and storm surges due to the impacts of climate change (Shahid, 2010). Estimates indicated that Bangladesh has about one fifth of the total area of the country which is affected by salinity and many people are already struggling for access to water resources for sustaining livelihoods, domestic uses, and health services (Rabbani et al., 2016). Salinity, both surface and groundwater, in the coastal zones of Bangladesh is really vulnerable to the health of local communities while the incidence of health vulnerability has become quite significant in those areas (Azam & Sarker, 2014).

Source : https://kus.ku.ac.bd/kustudies/article/view/186

IV. Agricultural change and livelihood stress

In coastal Bangladesh, livelihoods based on shrimp, fish and rice cultivation face growing risks from climate change-induced sea level rise, extreme flooding, cyclones, erosion and salinization.

In such conditions, crop diversification is considered an adaptation strategy for smallholder farmers to reduce vulnerability and enhance resilience. Although crop

diversification in Bangladesh remains low in the other parts of the country, it is on the rise in coastal areas, according to a recent study.

Studies reveal that surface water salinity remains high in these regions, and peaks from February to April, coinciding with the farming of summer-time crops.

During the post-monsoon season, farmers break the canals' connectivity to intertidal rivers with temporary embankments, restricting saline water intrusion into the crop fields. The reserved water is later used for irrigation in the dry season.

Source : https://news.mongabay.com/2024/10/coastal-farmers-in-bangladesh-give-up-shrimp-farming-for-agriculture-to-combat-salinity/

V. Development constraints for Khulna and Bangladesh

In recent years, Bangladesh was hit by two consecutive cyclones Sidr in 2007 and Aila in 2009. It was found that cyclone Sidr that hit Bangladesh on 15th of November 2007 caused about 3,406 deaths and over 55,000 people sustained physical injuries. Heavy rain accompanying cyclones and tidal waves due to wind effects caused extensive physical destruction, casualties, damage of crops and livestock, and flooding in a total of thirty districts across the South Western coastal district of Bangladesh. Cyclone Sidr affected nine districts of Bangladesh. The most devastated districts were Bagerghat, Barguna, Patuakhali, and Pirojpur.

For these reasons, scholars and policy analysts argue that tackling salinity and coastal vulnerability in Khulna must be treated as a national priority rather than a localised environmental issue, with urgent action required to prevent the gradual erosion of development gains achieved over recent decades.

Source : https://repository.uwl.ac.uk/id/eprint/3410/1/9654753%20-%20Second%20Round.pdf

Policy Recommendations

VI. 1. Climate-resilient and diversified agriculture

Agricultural research institutions in Bangladesh have released several salt-tolerant rice varieties and other crops designed for coastal conditions, yet adoption in many upazilas of Khulna Division remains uneven. Policy measures should include larger-scale participatory trials, farmer field schools and targeted input support (seed, fertilizer, extension visits) for smallholders to experiment with these varieties under local salinity regimes. Combining salt-tolerant *Aman* rice with short-duration pulses, oilseeds and vegetables in rotation can maintain soil productivity, distribute climate risk and improve both nutrition and income.

Source : https://www.geojournal.net/uploads/archives/7-6-8-277.pdf

VII. 2. Safe and sustainable drinking water

Entrepreneurship should be run with equitable and fair representation of all affected people. Moreover, alternative livelihood opportunities should be created by using locally available natural resources. In long term, small scale job opportunities may be introduced from the local investors, government and NGOs. Training on craft, tailoring, poultry and fish feed firming, ice factory, umbrellas' bat factory, match factory, cold storage and other small and medium scale industries may be introduced to alleviate the poverty and reduce vulnerability of the local people.

Damaged embankments and communication system should be repaired, and where necessary reconstructed, as priority basis to protect the affected area from further flooding and tidal surges. Height of the embankments should be increased together with proper afforestation with suitable tree species at both sides of the embankments. The relief programme should be run through ensuring equity and in a well coordinated way until the affected people can recover their damage. Government's VGF programme should focus the affected area as priority area. Moreover, monitoring of the relief distribution activities should be enhanced through the public participation and resources mobilization.

Source : https://unnayan.org/wp-content/uploads/2021/05/Cyclone-Aila-One-Year-on-Natural-Disaster-to-Human-Sufferings.pdf

VIII. 3. Climate-proof embankments and adaptive polders

The Bangladesh Water Development Board (BWDB) is responsible for maintaining and conducting the rehabilitation project of the polders. The BWDB has categorized 51 and 55 polders as most vulnerable and medium vulnerable, respectively. To address the vulnerability, it is important to repair the faulty infrastructure. Most of the sluice gates are damaged or broken which allow intrusion of saline water. Not only that, the shrimp farmers cut the embankment to get saline water into their shrimp fields which make the embankment weak. The weak embankment is easily damaged by tidal pressure, particularly during full moon, and thus saline water enters the polders. Regular maintenance of the embankments, dams, sluice gates, and polders are important to maintain the water balance. For this a robust monitoring system can be developed to ensure the accountability of the concern institutes and officials.

IX. 4. Livelihoods, social protection and gender-responsive adaptation

X.

Salinity-driven livelihood stress has pushed many coastal residents to migrate seasonally or accept precarious low-wage jobs. Programs that support small enterprises in non-farm sectors, such as, eco-tourism linked to the Sundarbans, processing of high-value crops (sunflower, watermelon, fruits) and handicrafts can reduce dependence on a single climate-sensitive livelihood like shrimp farming. Skills training and market-linkage support are important to ensure these activities are viable and inclusive for women and youth.

Evidence from Aila-affected areas shows that poor households often sell productive assets or incur high-interest debt to cope with disaster impacts, trapping them in prolonged poverty.

Women in coastal Khulna are central to household water management and caregiving yet often excluded from decisions on land, water and disaster planning. Building gender-responsive adaptation projects, policies should mandate women's representation in water user associations, disaster management committees and local development planning forums.

Source : https://www.thedailystar.net/opinion/editorial/news/ensure-safe-water-khulna-coast-3278721

XI. 5. Governance, data and participatory planning

Shrimp farming in the southwestern coastal areas was driven by high market demand in western countries. However, the government of Bangladesh while encouraging shrimp farming overlooked sustainability and planning issues, which has further aggravated the problems (Ahmed and Ambinakudige, 2023; Azad et al., 2009; Chowdhury et al., 2006). The lack of effective policies and coordination

among different organizations has hindered the government's ability to control the expansion of unplanned shrimp farming (Afroz and Alam, 2013; Deb, 1998). This has resulted in individuals seizing the opportunity to construct new shrimp ponds without considering environmental and societal concerns. Many of these ponds are located near agricultural areas and mangrove forests, accelerating the intrusion of soil salinity into non-saline regions (Rahman et al., 2010). Another concerning issue is the control of government-owned barren land in the southwestern coastal areas by local political leaders and influential individuals, who often convert it into shrimp ponds.

Source :
https://www.sciencedirect.com/science/article/abs/pii/S0964569124002266

Conclusion

Climate-induced salinity intrusion, compounded by recurrent cyclones and chronic waterlogging, is one of the most serious obstacles to sustainable development in Khulna Division, undermining agriculture, safe water, health and livelihoods and threatening Bangladesh’s wider progress toward food security and climate-resilient growth. Addressing this challenge requires an integrated package of measures: climate-resilient and diversified agriculture, safe and sustainable drinking-water systems, climate-proofed and adaptive embankments, stronger livelihood and social protection support, and cross-sectoral, gender-responsive governance rooted in community participation. Implemented together and tailored to local conditions, these strategies can help Khulna move from crisis-driven coping toward long-term resilience and inclusive development in a changing.